



Document type: **Infrastructure Upgrade Motivation**

Project Name: **Application Rationalization**

Created by	Thozamile Msitshana
Reviewed by	Nosizwe Mkwanazi, Nkaiseng Malete, Mmaditsela Letsoalo, Sithembiso Chivambo, Lulama Bikani
Quality assured	Nosizwe Mkwanazi, Nkaiseng Malete, Mmaditsela Letsoalo, Sithembiso Chivambo, Lulama Bikani
Approved by	Thandiswa Zide
Version	1
Date	10/03/2025



The Future. Reimagined.



Document History

Amended by	Version	Date	Reason for changes
Thozamile Msitshana	1	10/03/2025	Initial document

Document Management

Sign-off

The person/s listed below are responsible for signing-off the document, after careful consideration of its contents. The signed document will be accepted as knowledge, approval of the project and confirmation that this document is true and correct. It is the responsibility of the signatory/ies to ensure that all impacted stakeholders have a line of sight over this project. It is also up to the signatory/ies to raise any risks, omissions and or errors relating to the contents of this document, before signing.

Name	Department	Role	Signature	Date
Zide, Thandiswa	Solution Delivery	Business/Solution Owner		

Distribution

The person/s listed below are furnished with this document for information purposes only in order to obtain and or maintain line of sight of the implementation this project. Signing this document confirms knowledge of this project and its requirements. It is up to the signatory/ies to raise any risks, omissions and or errors relating to the contents of this document, before signing.

Name	Department	Role	Signature	Date
Zide, Thandiswa	Solution Delivery	Business/Solution Owner		

1. Background

The current infrastructure operates on a single-server architecture, leading to performance bottlenecks, limited redundancy, and scalability challenges. The system struggles with handling increased traffic, resource allocation, and fault tolerance, making an infrastructure upgrade and redesign necessary.

To address these challenges, the existing standalone servers will be repurposed as redundant servers, while the infrastructure will be enhanced with the following new components:

- Two Load Balancer Servers (for internal and external/DMZ traffic distribution).
- One API Gateway Server to centralize API management, security, and traffic routing.
- One backend server that runs all the backend services.

2. Current vs. Proposed Resources

Resource	Current Setup	Proposed Upgrade
Application Servers	Single application server (high risk of failure)	Multiple application servers behind a load balancer for redundancy and load distribution
API Management	Direct API exposure with security risks	API Gateway for centralized security, authentication, and traffic routing
Traffic Distribution	No load balancing, causing uneven server utilization	Load Balancer for optimized resource distribution and high availability
Security	Limited security mechanisms per service	API Gateway with JWT authentication, rate limiting, and traffic monitoring
Monitoring & Logging	Basic logging with limited visibility	Centralized logging & monitoring via API Gateway & Load Balancer

3. Current vs. Expected Hardware Resources

Current

Current Server	Current CPU	Current RAM	Current Disk Storage
152.106.2.172	4 vCPUs	16GB RAM	149GB
152.106.2.111	4 vCPUs	8GB RAM	49.4GB
152.106.2.54	4 vCPUs	8GB RAM	149GB
152.106.2.54	4 vCPUs	8GB RAM	149GB
152.106.2.54	4 vCPUs	8GB RAM	119GB
15.106.6.40	4 vCPUs	8GB RAM	80.5GB
15.106.6.40	4 vCPUs	8GB RAM	549GB

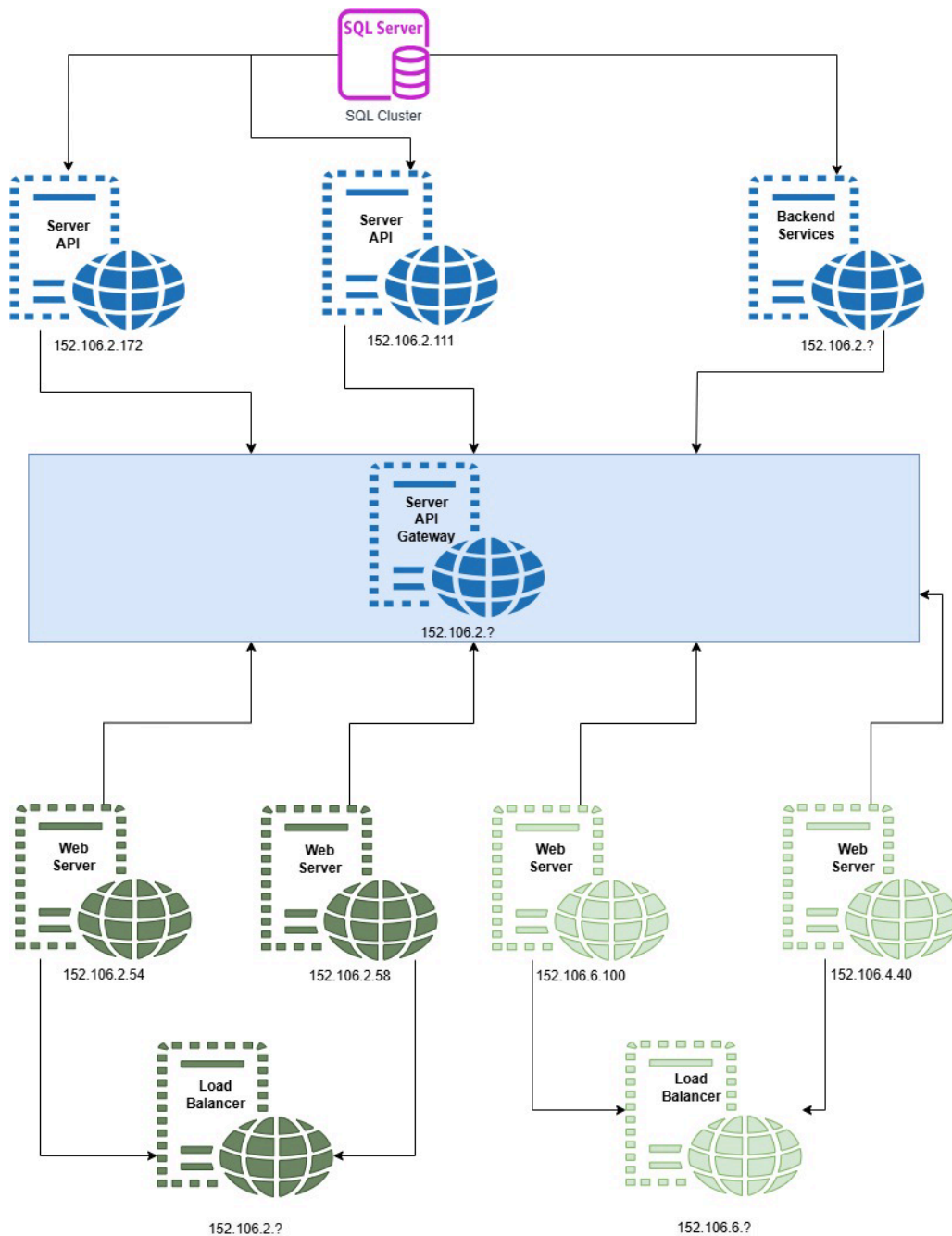
Expected

Resource	Expected Setup	Justification
CPU	8 vCPUs Minimum	Increased processing power for handling more concurrent requests
RAM	16GB RAM Minimum	Ensures smoother operation and better caching capabilities
Disk Storage	600GB Minimum	Supports larger datasets and high-performance read/write operations

4. Proposed Infrastructure Architecture

A new infrastructure architecture diagram will illustrate the upgraded system, including:

- Redundant application servers managed by a load balancer.
- Internal and external load balancers to distribute traffic efficiently.
- An API Gateway to enhance security, authentication, and request routing.
- A Backend application server to enhance the processing of backend jobs.



5. Key Benefits of the Upgrade

a. High Availability & Redundancy

- ✓ Eliminates single points of failure by using multiple servers.
- ✓ Load balancers distribute traffic efficiently, reducing downtime.
- ✓ Database clustering ensures data availability and failover protection.

b. Enhanced Performance & Scalability

- ✓ Increased CPU and RAM allow for better request handling and faster processing.
- ✓ Load balancing prevents server overload, improving response times.
- ✓ Auto-scaling dynamically allocates resources based on real-time traffic demand.

c. Improved Security & API Management

- ✓ API Gateway centralizes authentication and authorization, reducing security risks.
- ✓ Rate limiting and request filtering prevent DDoS attacks and API abuse.
- ✓ Improved monitoring and logging provide real-time visibility into API traffic.

6. Conclusion

Upgrading the infrastructure with a Load Balancer and API Gateway, alongside enhanced CPU, RAM, and disk resources, will significantly improve performance, scalability, redundancy, and security. This investment will future proof the system, ensuring seamless operations, minimal downtime, and optimized resource utilization.